

Polynomial interpolation and divided differences Let f be sufficiently smooth, and let $x_0, \dots, x_n$ be distinct real numbers.

1. **Symmetry of divided differences.** Prove that the n th divided difference is invariant under permutations of its arguments:

$$f[x_0, x_1, \dots, x_n] = f[x_{\sigma(0)}, x_{\sigma(1)}, \dots, x_{\sigma(n)}]$$

for every permutation σ of $\{0, \dots, n\}$.

2. **Recurrence formula.** Starting from

$$f[x_i] = f(x_i),$$

prove the recursive formula

$$f[x_i, \dots, x_{i+k}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i}, \quad k \geq 1.$$

Explain how this recurrence leads to an efficient construction of the Newton interpolation polynomial.

3. **Interpolation remainder.** Let p_n be the polynomial of degree at most n interpolating f at $x_0, \dots, x_n$. Prove that, for $x \notin \{x_0, \dots, x_n\}$,

$$f(x) - p_n(x) = f[x_0, \dots, x_n, x] \prod_{j=0}^n (x - x_j).$$

Then, assuming $f \in C^{n+1}$, prove the Lagrange form of the interpolation remainder,

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{j=0}^n (x - x_j),$$

for some ξ lying in the smallest interval containing $x, x_0, \dots, x_n$. (For this second question, it is possible to have $x = x_i$.)

4. **Optimal interpolation nodes.** Let

$$E_n(x) = \prod_{j=0}^n (x - x_j), \quad x \in [-1, 1].$$

Show that among all choices of $n+1$ distinct nodes $x_0, \dots, x_n \in [-1, 1]$,

$$\min_{\{x_j\}} \max_{x \in [-1, 1]} |E_n(x)|$$

is attained by the Chebyshev nodes

$$x_j = \cos\left(\frac{(2j+1)\pi}{2n+2}\right), \quad j = 0, \dots, n.$$

Determine the minimum value.

5. **The Runge function and interpolation error.** Consider the Runge function

$$f(x) = \frac{1}{1 + 25x^2}, \quad x \in [-1, 1].$$

- (a) Using a partial fraction decomposition over $\mathbb{C}$, derive an explicit expression for $f^{(n)}(x)$.

Hint. Factor the denominator over $\mathbb{C}$:

$$1 + 25x^2 = 25 \left(x - \frac{i}{5} \right) \left(x + \frac{i}{5} \right).$$

Using partial fractions, show that

$$f(x) = \frac{1}{2} \left(\frac{1}{1 - 5ix} + \frac{1}{1 + 5ix} \right).$$

You may then differentiate each term separately.

- (b) Use your expression to show that

$$\max_{x \in [-1, 1]} |f^{(n)}(x)| \leq n! 5^n.$$

- (c) Suppose first that p_n interpolates f at the Chebyshev nodes

$$x_j = \cos \left(\frac{(2j+1)\pi}{2n+2} \right), \quad j = 0, \dots, n.$$

Using the interpolation remainder formula, derive an upper bound for

$$\|f - p_n\|_{\infty, [-1, 1]}.$$

- (d) Now suppose that p_n interpolates f at the equally spaced nodes

$$x_j = -1 + \frac{2j}{n}, \quad j = 0, \dots, n.$$

Again, use the interpolation remainder formula to derive an upper bound for

$$\|f - p_n\|_{\infty, [-1, 1]}.$$

- (e) Compare the two bounds as n increases. What does this calculation suggest about the choice of interpolation nodes for the Runge function?